INDEPENDENT WHITEPAPER · PAKISTAN LIVESTOCK & POULTRY

The Smart Livestock Asset Stack

From biometric passport to exchange rails — a farm-to-market design for digitizing, certifying, tokenizing, and pricing animal assets.

Independent · Ammar Jamshed

Author: Ammar Jamshed (personal capacity)

Version: 1.0 · 2026

Site: https://ammarjamshed.github.io/livestock-smart-asset-stack-whitepaper/

GitBook: https://livestockmon.gitbook.io/smart-livestock-asset-stack/

Stance: No institutional affiliation, endorsement, or mandate

SMART LIVESTOCK ASSET STACK

Cover & how to use

The Smart Livestock Asset Stack

From the barn gate to the exchange board

Farm → Registry → Health → Token → Market grid
Independent research by **Ammar Jamshed** — no bank, exchange, ministry, or vendor allegiance.

Primary reading site (livestock ops UX): ammarjamshed.github.io/livestock-smart-asset-stack-whitepaper

Also on GitBook: livestockmon.gitbook.io/smart-livestock-asset-stack · Page Actions → **Export as PDF**

Farm metaphor	Stack layer	Job
Ear tag / muzzle scan	L0 Field	Animal Passport / PAITS capture
Title deed for the animal	L1 Registry	Legal company system of record
Vet desk grade	L2 Health	Certified daily assessment feeds
Share of the herd / flock	L3 Token	PVARA-aligned mint gates
Mandi board → national board	L4 Grid	Daily value feeds → PSX / CDC



Walk the farm

If you want…	Open this gate
The one-page claim	Executive summary
Why animals stay off-book	The problem

If you want…	Open this gate
Field apps (Animal Passport)	Field identity
Vet / health rail	Health assessment
Full yard map	Architecture & flows
Build sequence	How to achieve it
Author + independence	About · Disclaimer
PDF	Download PDF

Institutions named in this paper (PSX, CDC, PVARA, AKUH, MoH, Animal Passport, PAITS) are **proposed counterparties or public references** — not signed partners.

Suggested citation

> Jamshed, A. (2026). *The Smart Livestock Asset Stack: From Biometric Passport to Exchange Rails in Pakistan* (Independent whitepaper, v1.0).

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Executive summary

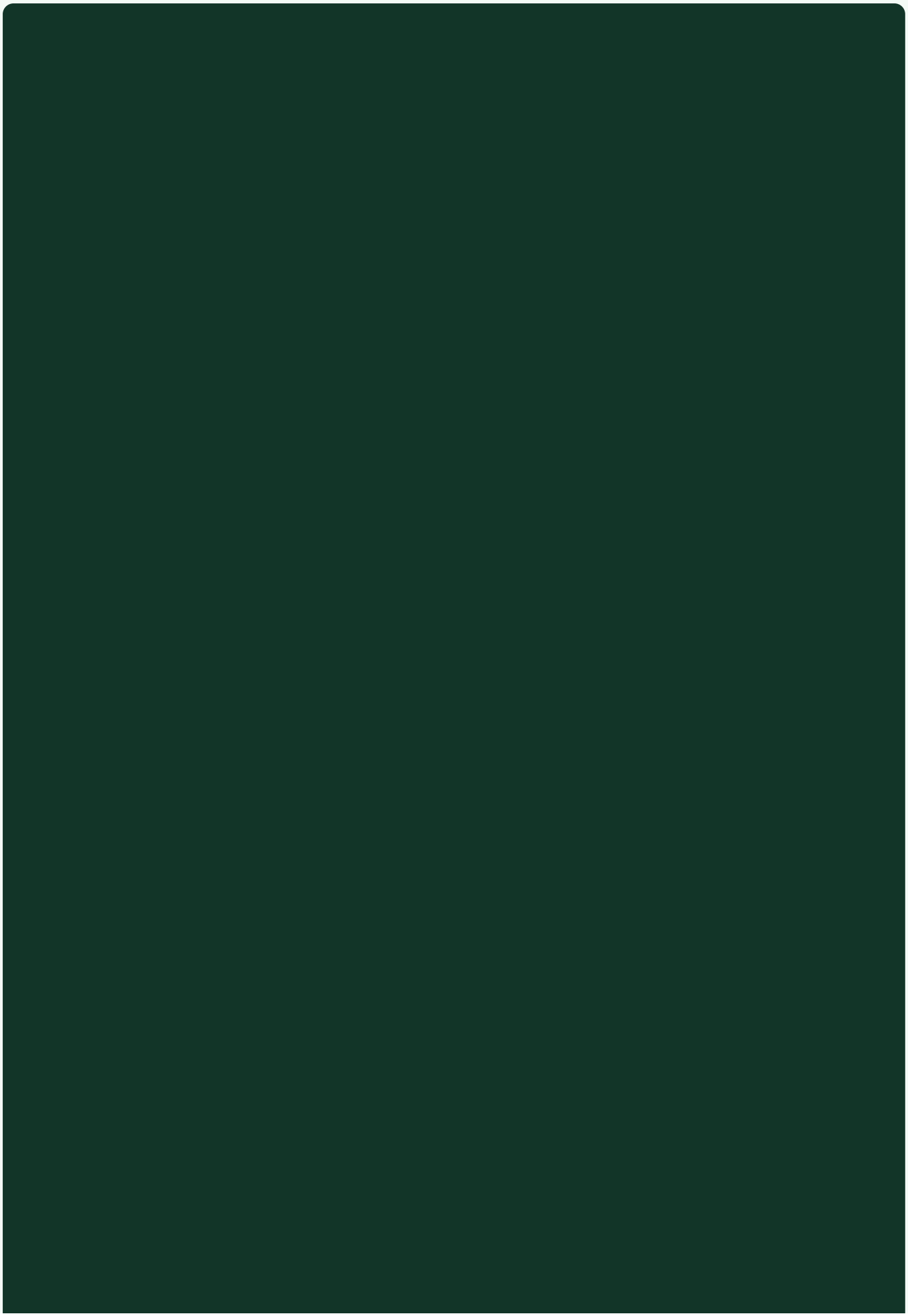
Executive summary

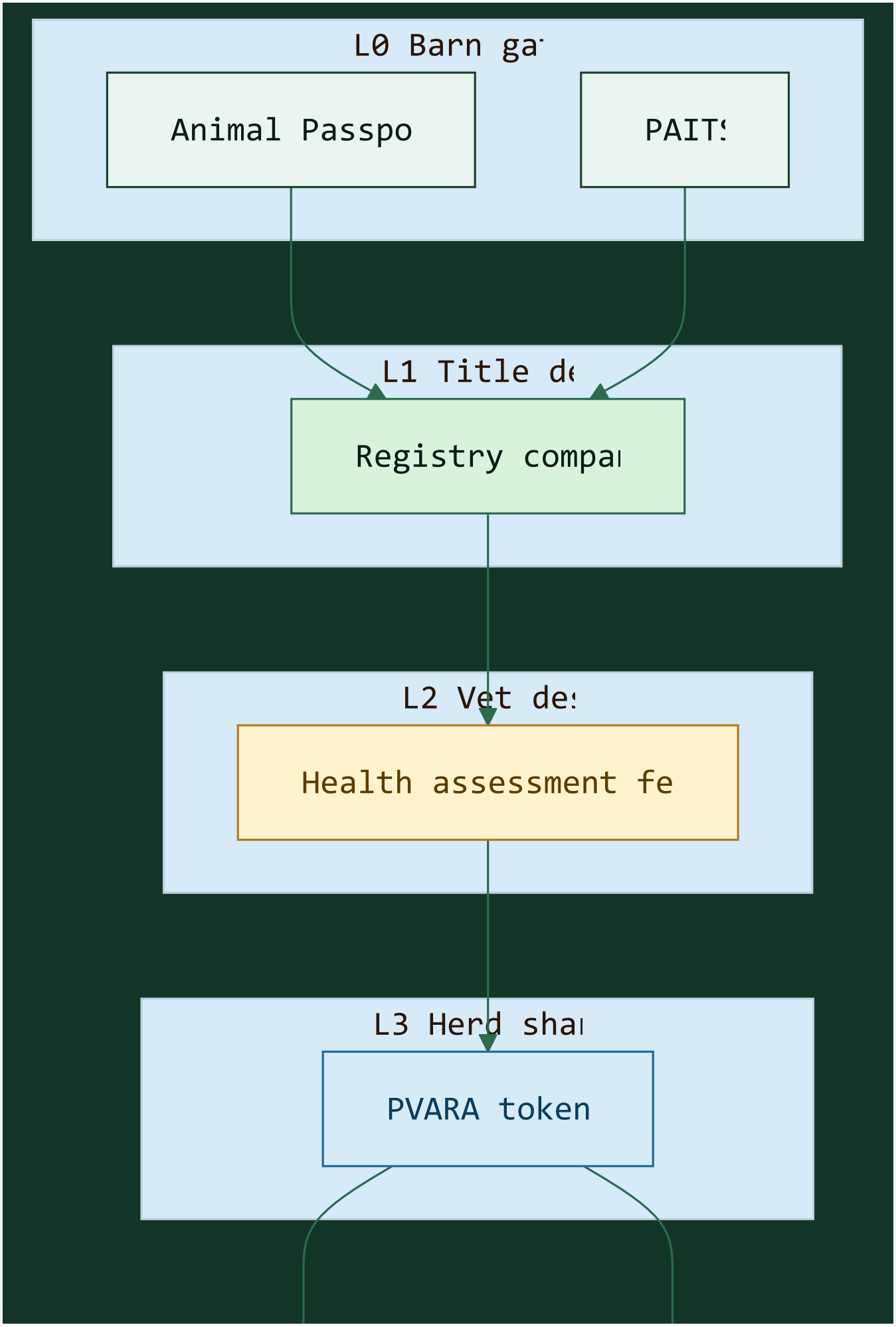
One-line claim: Pakistan already has farm-gate digital identity — what is missing is a **barn-to-board stack** that turns it into bankable, attestable, market-visible assets.

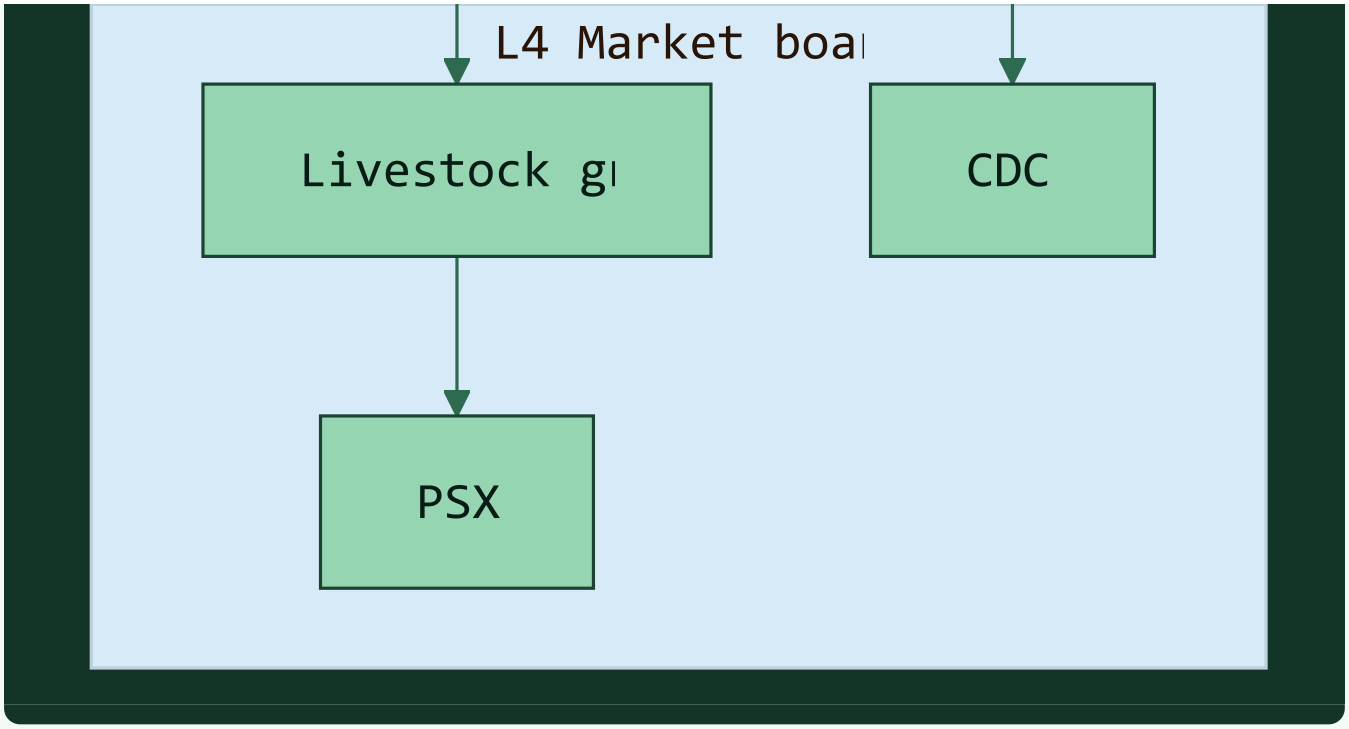
The smart process (SLAS)

SLAS — Smart Livestock Asset Stack — five farm-to-market layers:

Layer	Farm metaphor	Job
L0 Field	Ear tag / muzzle scan	Animal Passport; PAITS
L1 Registry	Title desk	Legal system of record; CDC sub (phased)
L2 Health	Vet desk	Certified daily grades
L3 Token	Herd / flock share	PVARA-aligned mint
L4 Grid	Market board	Daily value feeds → PSX / CDC




L0 Barn ga
Animal Passpo
PAITS
L1 Title de
Registry compa
L2 Vet des
Health assessment fe
L3 Herd sha
PVARA token



Why this is novel

Apps, traceability pilots, exchanges, and virtual-asset talk already exist. **Novelty = wiring them into one farm-to-market process.**

What success looks like

- Verified animal **promoted** into a registry title
- Title carries a **vet-desk health grade** on a defined clock
- Only passing stock underpins a **PVARA-compliant token**
- A **daily value board** publishes prices with verified-inventory %
- Banks/insurers price credit and cover off the stack — not notebooks

Not a product pitch, buy/sell call, or claim that named institutions have signed on.

SMART LIVESTOCK ASSET STACK

The problem

The problem

Broken fence: animal wealth is huge on the farm — but **off-book** for banks, insurers, and markets.

Three failures that keep animal wealth “off-book”

1. The ear tag washes away

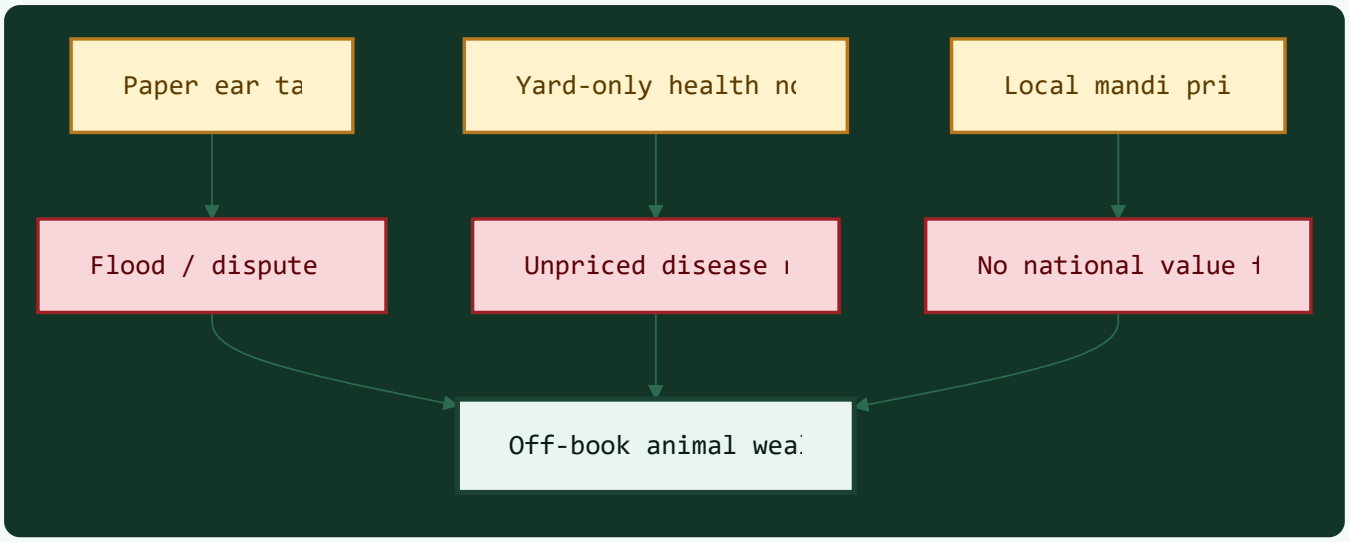
Paper tags, notebooks, and village memory collapse under floods, theft disputes, and cross-district sales. When water rises, **proof of ownership disappears with the paper**. Biometric and national-traceability tools exist because this pain is real — but they are not yet wired into capital markets.

2. The vet note never leaves the yard

A buyer, lender, or insurer rarely gets a **standardized, signed health feed** tied to a durable animal ID. Outbreak risk stays informal. Without portable health grades, insurance pricing and token mint gates cannot be honest.

3. The mandi board never becomes a national board

Live animal and poultry value is discovered in mandis and farm-gate deals that do not produce a **national daily value surface**. Capital markets see equities and bonds — not a clean livestock grid. Illiquidity is structural, not accidental.



Why “just build a farm app” is not enough

An app can register a cow. A **smart asset stack** must also answer title, access, attestation, mint, and price questions.

Question the stack must answer	Farm metaphor	Layer
Who is the **legal issuer of record**?	Title desk	Registry company
Who may **query** (bank, insurer, exchange)?	Who holds the keys	Access control
What upgrades a barn note into a market health feed?	Vet desk stamp	Health feeds
When may a **token** be minted or frozen?	Green lane / red pen	PVARA mint gate
How does a **price** get published safely?	Market board	Value grid

Design principle

Identity without custody is a database.
Custody without identity is fiction.
Price without both is noise.

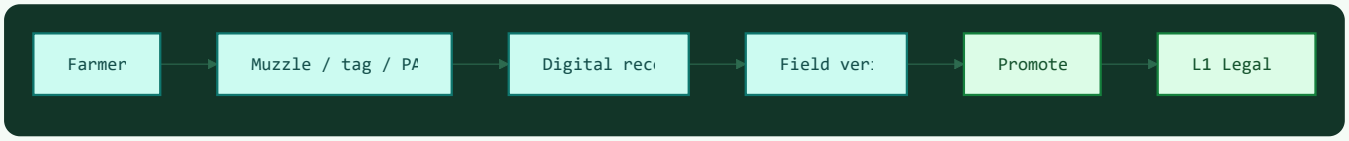
SLAS requires identity + health + token + market rails before claiming livestock is “on the grid.”

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Field identity: Animal Passport & PAITS

Field identity: Animal Passport & PAITS

Role: L0 — capture and verification. Not the registry, not the exchange, not the token issuer. This is how living animals **enter** the smart process.



Animal Passport — private product edge

Pakistan-rooted biometric livestock platform: muzzle/nose ID, owner binding, health/breeding history, APIs, flood-recovery identity.

SLAS need	How it assists
Unique living asset	Biometric ID + optional tag
Owner binding	Owner account + contacts
Disaster continuity	Cloud ID survives lost tags
Integration	Verification API
Health raw events	Farm history → L2 inputs

PAITS — public policy edge

PAITS (PITB + FAO for GoP): national ID, tracing, health check, monitoring — **government data gravity**. Map private biometric IDs to PAITS keys (dual-rail).

Integration models

Model	Idea	Prefer when
A Edge vendor	Apps onboard; registry is SoR	Fast multi-vendor pilot
B Dual-rail	PAITS + Animal Passport	Legitimacy + innovation
C Licensed operator	One capture agent / province	Deep MVP
D Standards-only	Any certified app	Open long-term market

v1 recommendation: Model **B (dual-rail)** — one province, one species.

Do **not** claim Animal Passport or PAITS are already wired to PSX, CDC, PVARA, or AKUH without a signed MoU.

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Registry company

Registry company

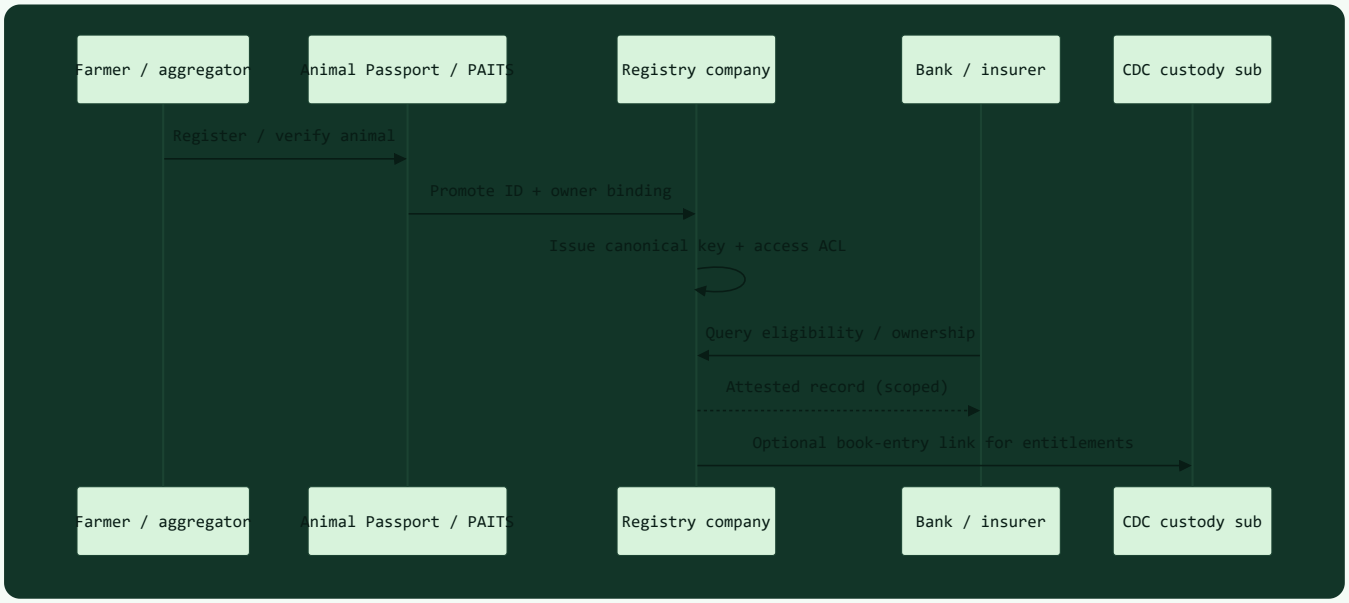
L1 — Legal spine. Your notes correctly struck “Access” and circled **company**. A query portal is not enough.

Notebook origin

Pakistan needs a **registry company** — a legal person that issues and maintains animal / batch identities, governs who may read or write them, and can interface with custody (CDC sub) and health rails.

Job of the registry

1. Accept **promoted** field IDs (Animal Passport / PAITS / peers)
2. Issue a **canonical registry key** (system of record)
3. Bind owner / aggregator KYC to that key
4. Expose **controlled APIs** to banks, insurers, token issuers, and (later) market data agents
5. Maintain dispute, re-tag, death, and transfer events



Design choices to settle in the paper’s pilot

Choice	Options	Lean for MVP
Legal form	Section company / utility / concession	Narrow-purpose company with clear public-interest charter
ID source of truth	Field app vs registry	**Registry** wins; field apps are edges
Access model	Open vs licensed queriers	Licensed queriers (banks, insurers, issuers)
CDC link	Subsidiary vs special accounts	Phased: accounts first, sub later if volume justifies
Poultry vs cattle	Per-animal vs batch IDs	Cattle: per-head; poultry: **batch / flock** IDs

Must-have data fields (minimum)

- Canonical registry ID
- Field keys (passport biometric ref, PAITS ID, ear tag)
- Species / breed / sex / approx. age or DOB estimate
- Owner / aggregator identifiers (CNIC / NTN as applicable)
- Status: active · transferred · dead · quarantined · disputed
- Last health grade pointer (to L2 feed)
- Audit log hash / version

Failure modes

- Double registration across apps → **merge rules** and biometric match thresholds
- Owner disputes → escrow + local verification workflow
- Ghost animals for credit → mandatory field re-verify before token mint or large loans

SMART LIVESTOCK ASSET STACK

Health assessment & daily feeds

Health assessment & daily feeds

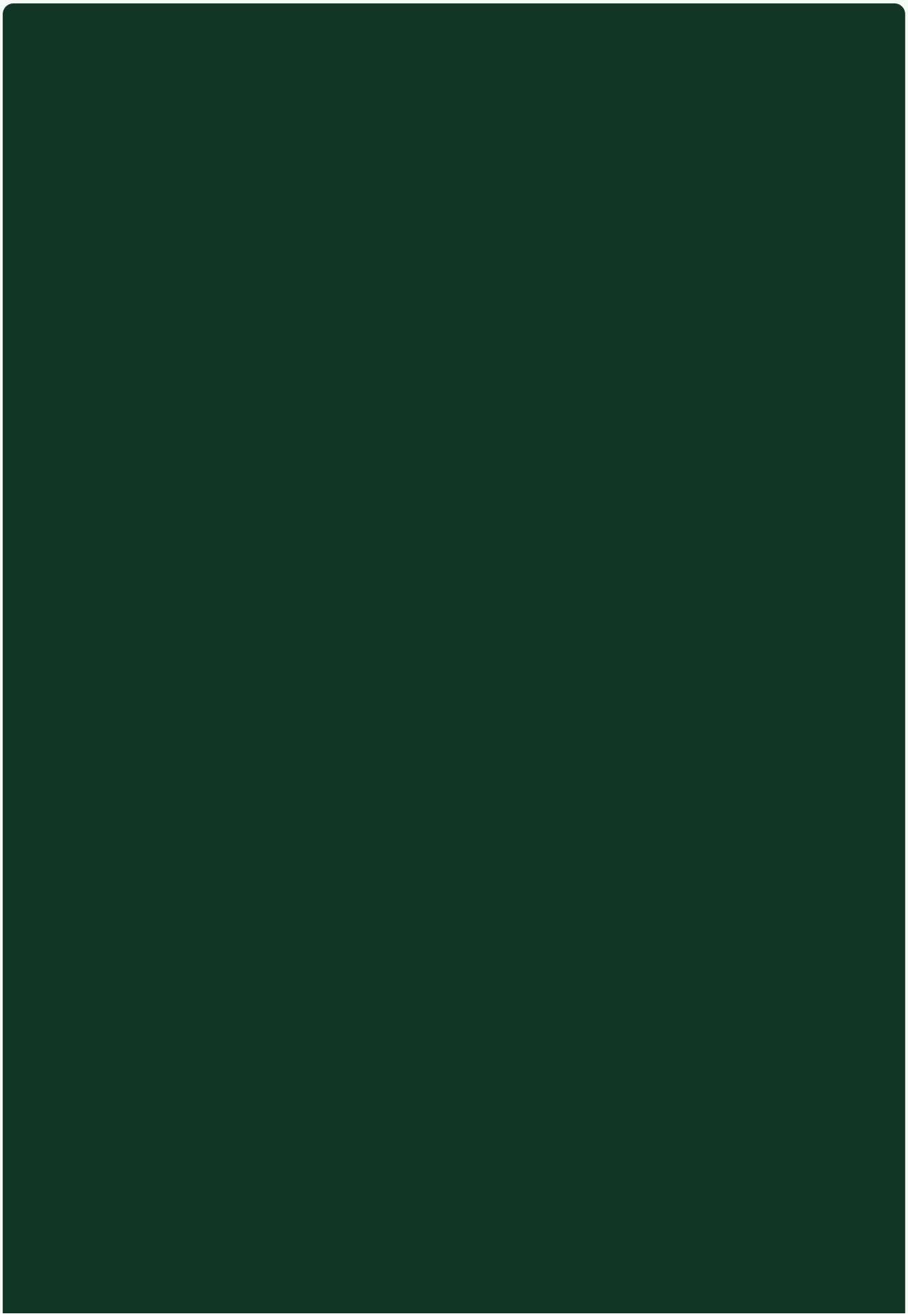
L2 — Farm health rail (amber / hay). Barn notes and AI scores are the *raw feed*. Markets need a signed daily health grade — like an ear-tag that banks and token gates can read.

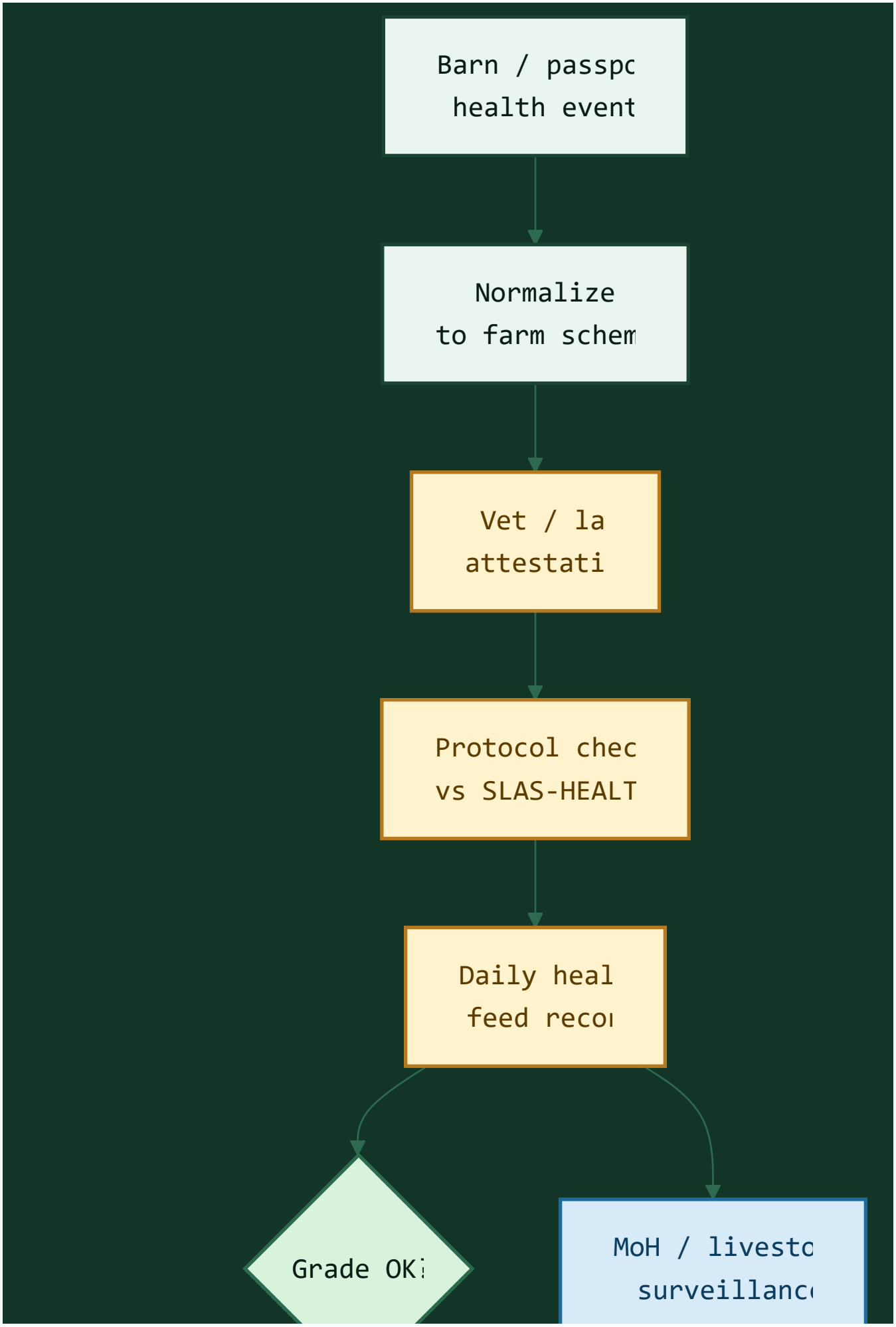
Role in the stack

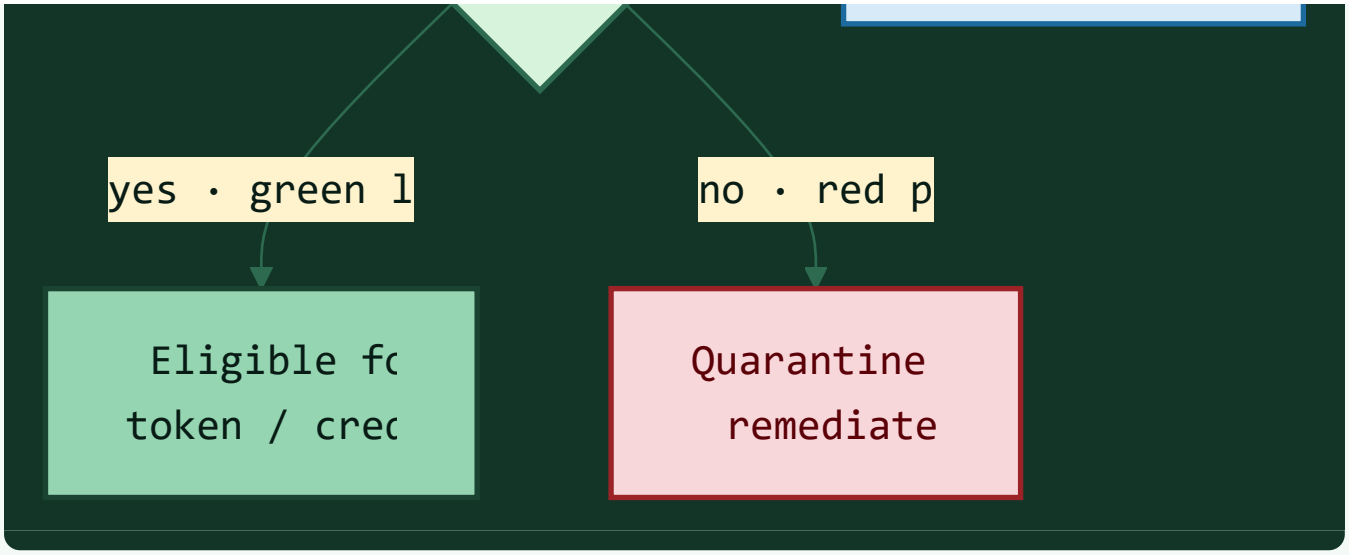
Think of this layer as the **vet desk between the paddock and the market**:

Farm side (inputs)	Market side (outputs)
Passport / farm-app health events	Attested **daily health feed**
Local vet & lab notes	Grade **A / B / C / Quarantine**
Flock mortality & feed flags	Freeze / mint eligibility signals

Proposed institutional path from the notebook: **clinical credibility** (e.g. AKUH standards path) plus **public-health policy** (Ministry of Health) with livestock / veterinary ministries. These are **design counterparts**, not claimed partners.




Barn / passpc
health event
Normalize
to farm schem
Vet / la
attestati
Protocol chec
vs SLAS-HEALT
Daily heal
feed reco
Grade OK
MoH / livesto
surveillanc



Green lane = grade passes → credit / token mint may proceed.
Red pen = quarantine → freeze transfers until cleared.

Two cadences (farm clock)

Cadence	What the farm sees	What the market uses it for
Daily / near-daily	Mortality, feed intake flags, flock alerts, movement	Outbreak alerts + token freeze
Periodic	Vaccination, lab panels, body condition, AI age/health scores	Insurance bands + mint eligibility

Feed record (ear-tag style stub)

```
registry_id          # the animal / flock "tag"
timestamp_utc
metrics[]            # coded barn observations
attestor_id          # vet / lab / protocol agent
attestor_sig
protocol_version     # e.g. SLAS-HEALTH-0.1
grade                # A / B / C / Quarantine
validity_until
source_refs[]        # passport event IDs, lab report IDs
```

Who does what on the farm vs in town

Actor	On the farm / yard	Does not
Animal Passport / farm apps	Capture raw signals & history	Issue market certificates alone
Local vets / labs	Attest what they see	Set national protocol alone
Standards partner (e.g. AKUH path)	Protocol, audit, training	Run every shed
MoH + livestock ministries	Outbreak / zoonoses rails	Replace farm capture

Why this layer matters: a token that cannot be frozen when herd health collapses is a fiction sold to investors. L2 is the ethical core of the stack — the difference between a living asset and a paper cow.

SMART LIVESTOCK ASSET STACK

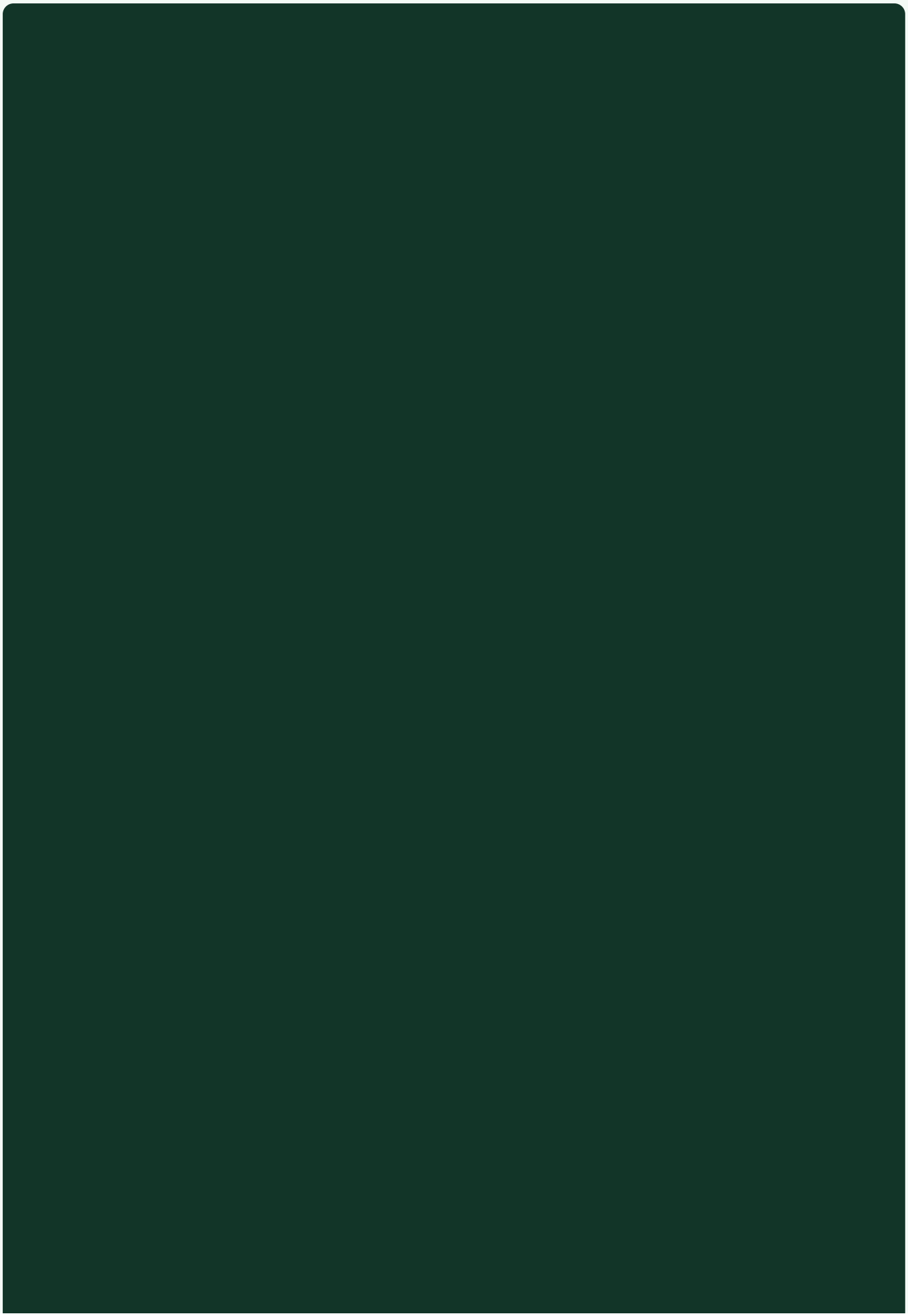
PVARA tokenization

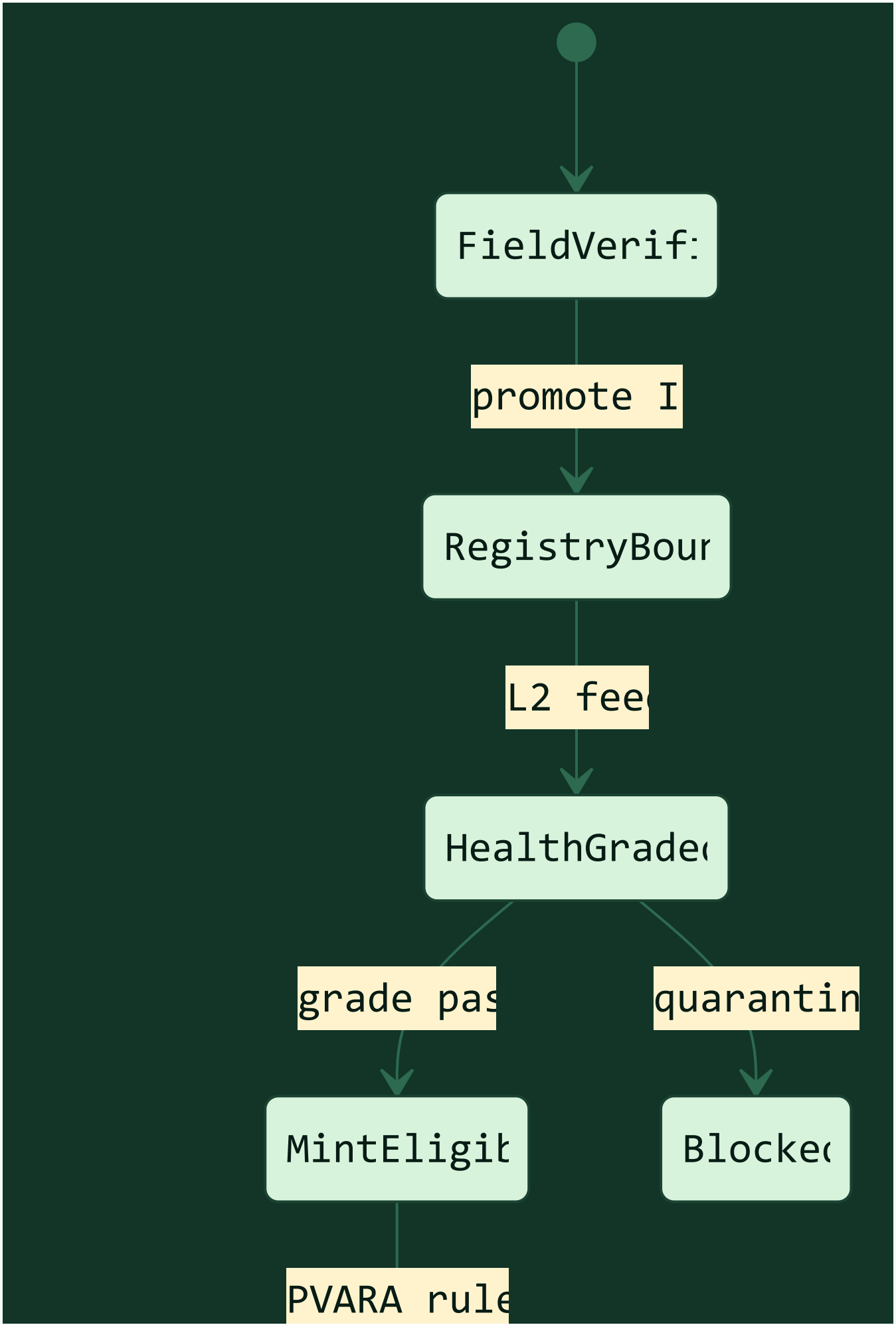
PVARA tokenization

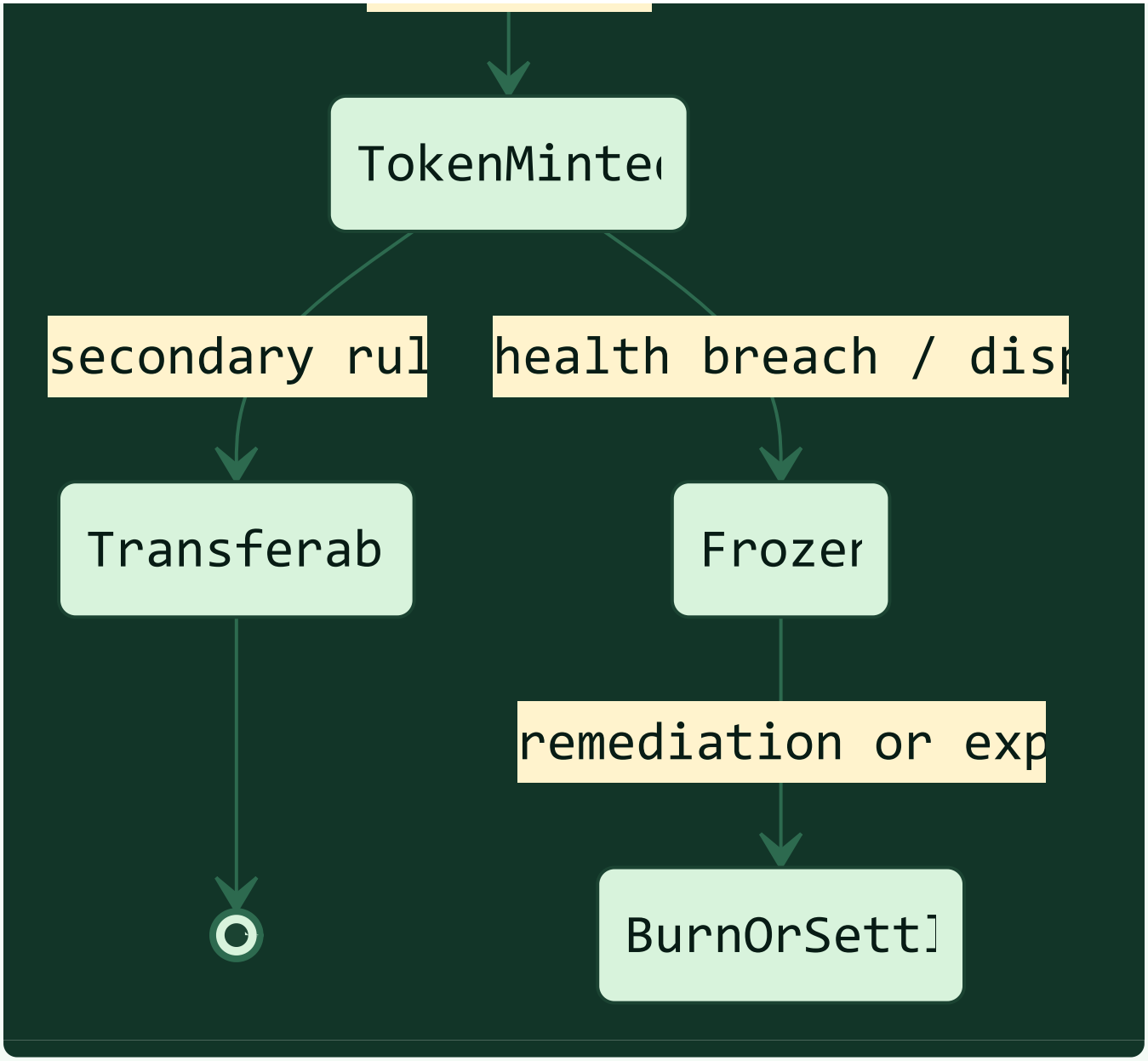
L3 — Sky layer. Tokens mint **only after** L1 identity + L2 health pass. Notebook: **PVARA → Livestock Tokenization & Poultry**.

Role in the stack

Tokenization creates transferable economic claims on verified living assets or batches — **only after** L1 identity and L2 health pass.




FieldVerif:
promote I
RegistryBour
L2 fee
HealthGrade
grade pas
quarantin
MintEligit
Blocke
PVARA rule



What the token can represent (pick explicitly)

Object	Fits	Caution
Single large animal economic claim	Breeding / fattening cattle	Welfare & delivery logistics
Flock / batch claim	Poultry cycles	Batch ID discipline
Pooled herd share	Diversified risk	NAV methodology
Cashflow claim (milk / eggs)	Shorter tenor products	Oracle dependence

Mint gate (non-negotiable)

```
MAY_MINT =
  registry.status == active
  AND health.grade in {A, B}
  AND health.validity_until > now
  AND no open dispute
  AND not already pledged beyond limit
```

Custody path

- **Phase 1:** issuer register + off-exchange transfer rules
- **Phase 2: CDC** book-entry entitlements (notebook: CDC sub)
- **Phase 3:** optional PSX-visible products once liquidity and disclosure mature

Compliance posture (independent framing)

This paper does **not** assert a final legal classification. The design assumes issuers will seek clarity under Pakistan's evolving virtual-asset / securities perimeter (**PVARA** and related authorities). Shariah review should be treated as a first-class workstream for farmer and retail acceptance.

Anti-abuse

- Biometric / PAITS re-verify before large mints
- One registry ID → controlled pledge graph (no silent double-count)
- Automatic freeze hooks from L2 quarantine flags

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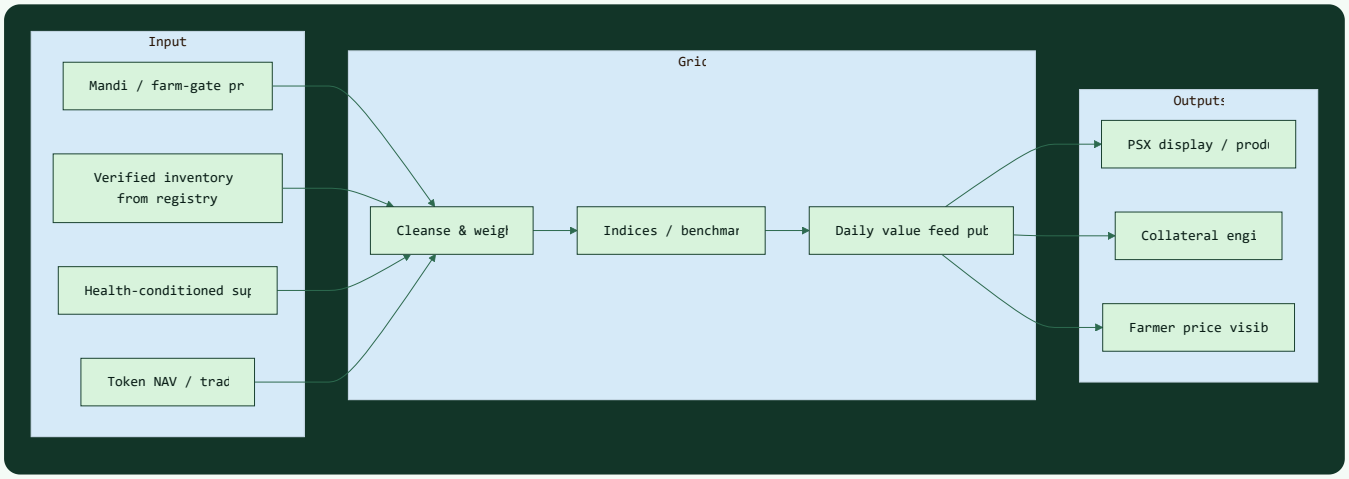
Livestock & poultry grid

Livestock & poultry grid

L4 — Forest layer. National aggregation into **daily value feeds**. Notebook: Grid → daily value feeds → PSX (+ CDC, phased).

Role in the stack

The grid turns fragmented farm-gate and mandi observations into **daily value feeds**. Listing and custody stay separable.



Product ladder (do not jump to listing day one)

Stage	Product	PSX role	CDC role
0	Private benchmark (pilot region)	None	None
1	**Information product** — daily published feed	Display / data partnership	Optional
2	Index / total-return style benchmarks	Branding / methodology review	None / light
3	Listed or admitted products	Market venue	Custody for entitlements

Notebook strike on “CDC” next to PSX is useful: treat **listing** and **custody** as separable. Publish prices before you clear tokens.

Daily value feed stub

```
instrument_code
region
unit
price
currency
methodology_version
sources[]
pct_verified_inventory
publish_time_pk
```

Integrity rule

A grid that ignores verification percentage will be gamed. Publish **`pct_verified_inventory`** beside every headline price so users see how much of the print sits on passport/registry-backed stock.

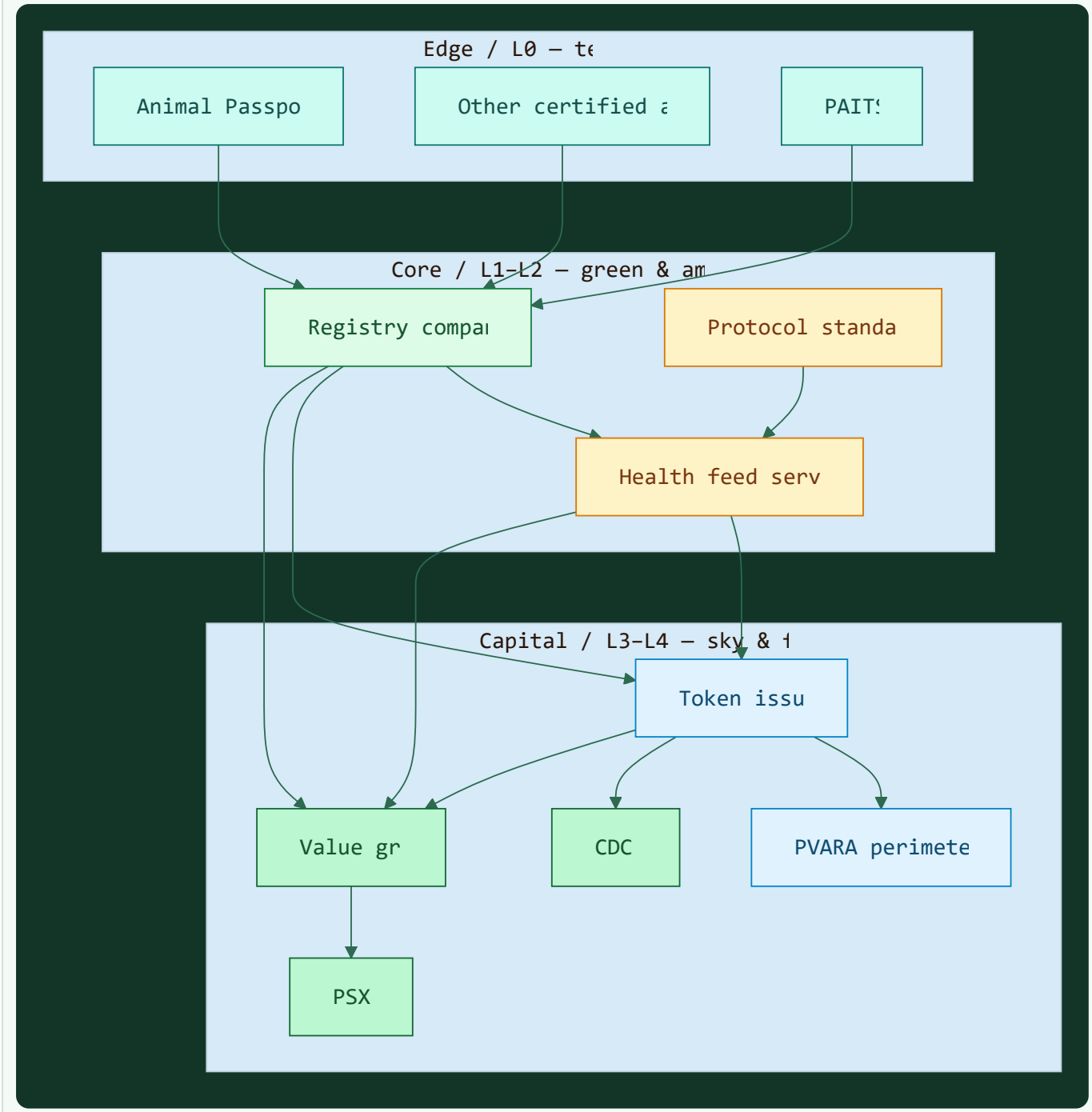
SMART LIVESTOCK ASSET STACK

End-to-end architecture & flows

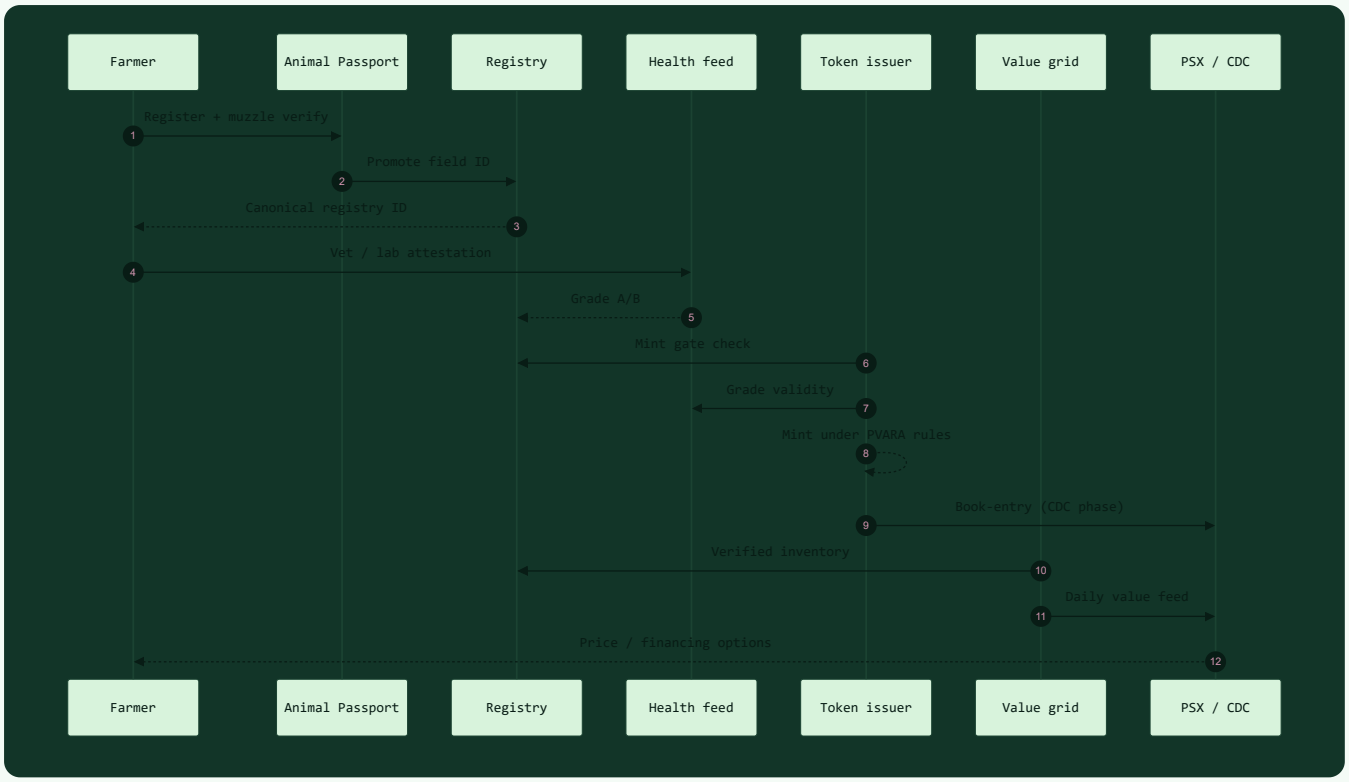
End-to-end architecture & flows

This chapter is the **visual spine** of the whitepaper — follow the coloured layers from farm to market.

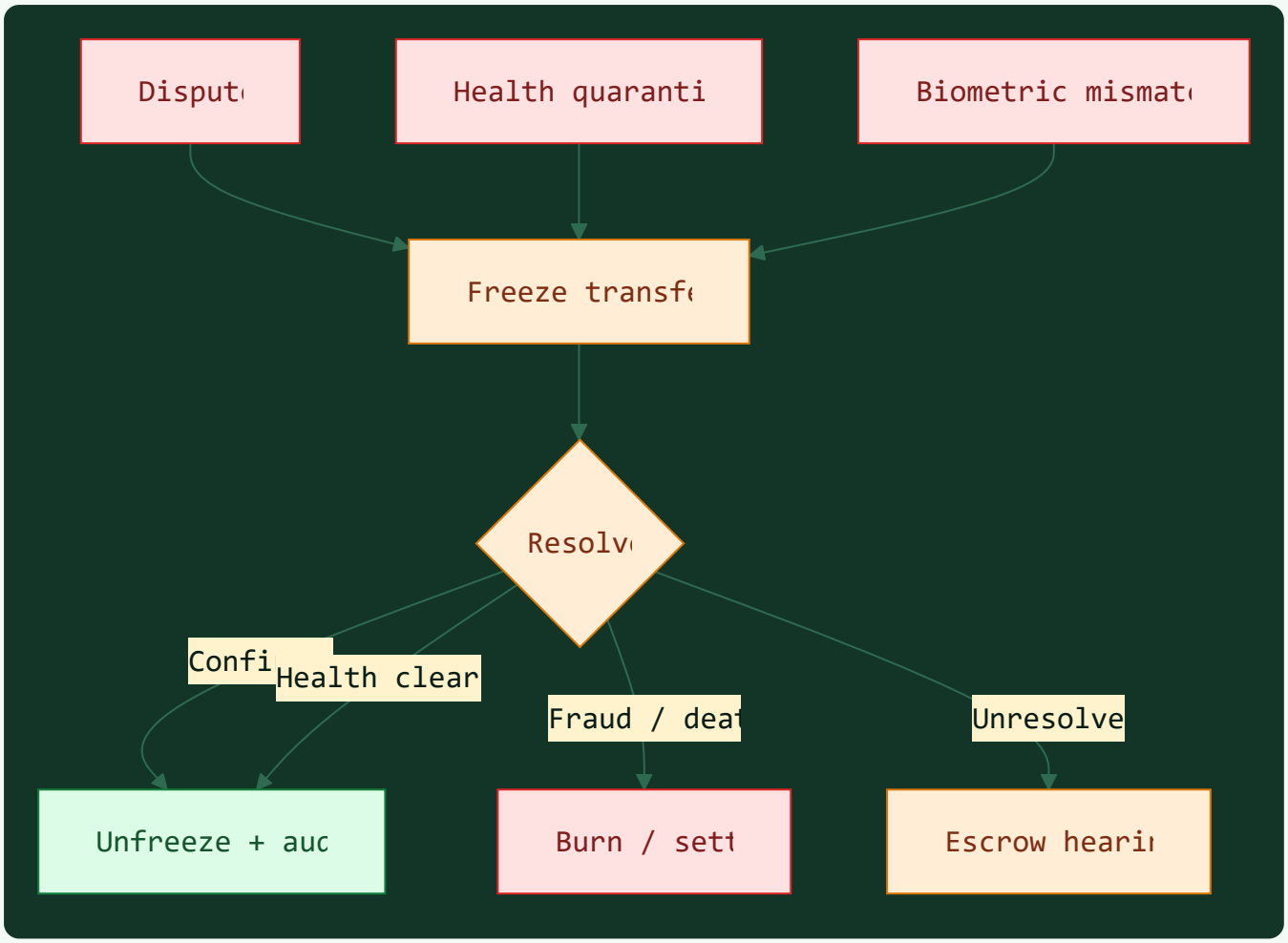
Layer map


Edge / L0 — te
Animal Passpo
Other certified a
PAITS
Core / L1–L2 — green & am
Registry compar
Protocol standa
Health feed serv
Capital / L3–L4 — sky & f
Token issu
Value gr
CDC
PVARA perimete
PSX

Happy path



When something goes wrong



Readiness checklist

Capability	Ready when…
Capture	≥1 certified edge app writing promotes
Legal ID	Registry live with ACL + audit log
Health	Protocol v0.1 + attestor network
Token	Mint gate enforced in code + docs
Grid	Daily feed with methodology note
Market	Stage-1 PSX info product **or** private benchmark used by a bank

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How to achieve it: roadmap

How to achieve it: roadmap

Build sequence for an independent convenor — not an institutional mandate.

This chapter turns the architecture into phases you can actually run.

Phase 0 — Paper & coalition (4–8 weeks)

Outcomes

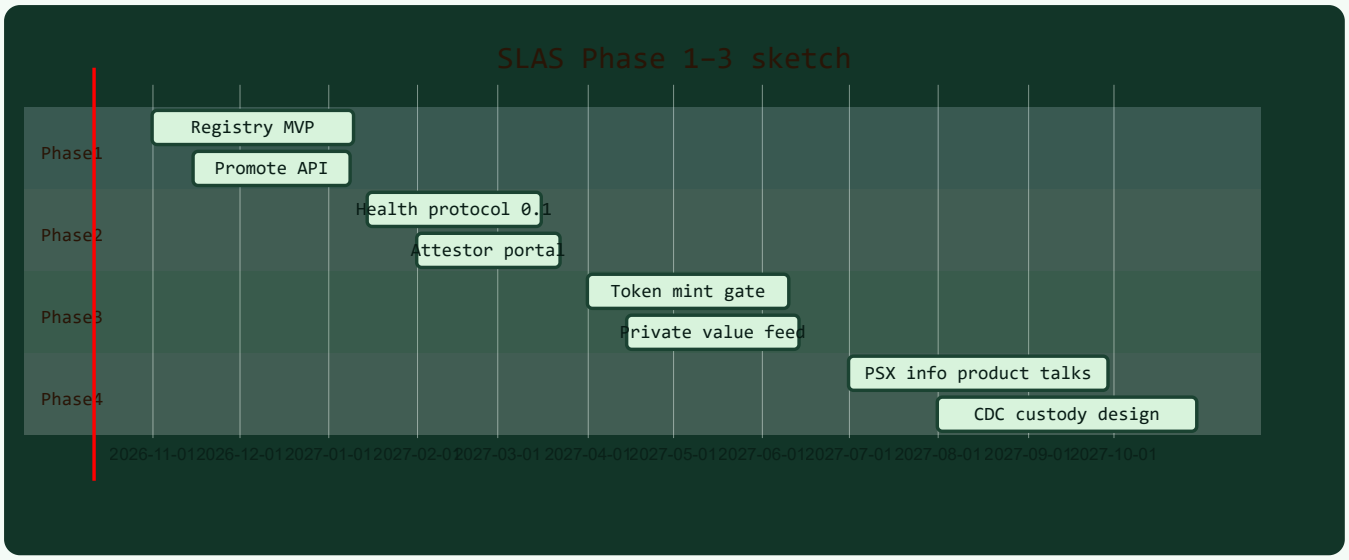
- Freeze SLAS vocabulary and chapter claims (this GitBook)
- Choose province + species (e.g. Sindh cattle **or** Punjab broiler batches)
- Choose integration model (**B — dual-rail** recommended)
- Draft one-pagers for registry company counsel, one edge platform, one bank, one insurer

Exit criteria: signed interest letters (non-binding) from ≥3 parties.

Phase 1 — Identity spine (8–16 weeks)

Build

- Registry company incorporation + data policy
- Promote API from Animal Passport-class edge (+ PAITS ID mapping where available)
- Owner KYC lite + dispute workflow
- Bank read API (eligibility only)



Exit criteria: 1,000+ promoted live IDs (or equivalent poultry batches) with <2% unresolved disputes.

Phase 2 — Health feeds (parallelizable, 8–14 weeks)

Build

- Publish `SLAS-HEALTH-0.1` schema
- Onboard attestors (vets / labs)
- Daily job that writes grades to registry pointers
- Freeze hooks for quarantine

Exit criteria: ≥70% of pilot inventory has a non-expired grade.

Phase 3 — Controlled tokenization (10–16 weeks)

Build

- Legal memo on instrument type under PVARA perimeter
- Mint gate implementation (identity + health + pledge graph)
- Investor disclosures + Shariah review track
- Optional internal transfer register

Exit criteria: successful test mint/burn cycle on **sandbox** assets with zero double-count incidents.

Phase 4 — Grid & market visibility (12–20 weeks)

Build

- Private daily value feed for pilot region
- Publish `pct_verified_inventory`

- Bank collateral simulation
- Conversations with PSX for **Stage-1 information product** (not rushed listing)
- CDC custody design workshop (phased)

Exit criteria: 60 consecutive publish days; at least one financial institution consumes the feed in a pilot credit or insurance process.

RACI (starter)

Workstream	Responsible	Accountable	Consulted	Informed
Registry company	Incorporators / ops lead	Board of registry co.	Counsel, edge apps	Farmers
Field capture	Edge platform	Registry co.	PAITS/PITB	MoU parties
Health protocol	Standards working group	Protocol chair	Vets, MoH, livestock dept	Issuers
Token rules	Issuer + counsel	Issuer board	PVARA perimeter advisors	Investors
Value feed	Grid methodology lead	Grid steward	Mandi agents, banks	PSX (when engaged)
Custody	CDC design counterparts	Registry / issuer jointly	SECP-related advisors	Token holders

Independent convenor model

Because this paper is **not** an institutional project, achievement depends on a **light convenor**:

1. Publish the blueprint (this GitBook + PDF)
2. Host a closed workshop with edge, bank, insurer, counsel
3. Spin the registry company as a neutral utility
4. Keep vendors interchangeable via schemas and certification

Budget & talent sketch (order-of-magnitude)

Bucket	Notes
Legal / company setup	Incorporation, data protection, MoUs
Engineering	Registry, promote API, attestor portal, feed publisher
Field ops	Onboarding, verification drives, dispute desk

Bucket	Notes
Clinical protocol	Standards writing, training, audits
Market design	Methodology, disclosures, PSX/CDC conversations

Exact PKR figures should be built bottom-up for the chosen province; do not fake precision in v1.

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Risks, open questions & research agenda

Risks, open questions & research agenda

Material risks

Risk	Why it matters	Mitigation
Vendor capture	One app becomes de facto registry	Registry as system of record; certify multiple edges
Fake inventory	Tokens on ghost animals	Biometric re-verify; pledge graph; attestor liability
Health theater	Grades without audits	Protocol + surprise audits + freeze hooks
Premature listing	PSX brand used before methodology mature	Stage-1 information products first
Regulatory fog	Token class unclear	Sandbox + counsel; no retail until clarity
Farmer exclusion	Digitization without access	Free/low-cost capture; aggregator model; Urdu UX
Zoonosis politics	MoH vs livestock turf	Explicit dual-ministry RACI

Open research questions

1. What biometric FAR/FRR is acceptable for **credit-scale** mint gates in Pakistani field conditions (wet muzzle, dust, low light)?
2. How should **poultry batch IDs** inherit and shed birds without breaking token accounting?
3. Can PAITS keys and private biometric IDs be linked with privacy-preserving joins?
4. What is the minimum disclosure pack for a PVARA-perimeter livestock token?
5. Which PSX product type (data, index, listed) maximizes farmer benefit vs investor confusion?
6. How do Islamic finance structures map onto fattening cycles and flock turns?

What v2 of this whitepaper should add

- Primary interviews (edge platform, PITB/PAITS, bank agri desk, insurer)
- A filled cost model for one district
- A worked example: 500 cattle fattening cycle through mint → feed → settle
- Comparative annex (India / EU cattle ID, commodity token precedents) with citations

Honest limit of v1

This document is a **design whitepaper**, not an empirical evaluation and not a prospectus. Its job is to make the smart process legible enough that practitioners can agree — or disagree — in public.

SMART LIVESTOCK ASSET STACK

About the author

About the author

Ammar Jamshed — independent researcher & writer. This whitepaper is personal work only.

Ammar Jamshed

Focused on data, market infrastructure, and practical digitization problems in Pakistan's real economy.

Institutional names in the author's background are **context only** — not sponsors, employers-of-record for this work, co-authors, or endorsing bodies.

Background (context only)

Ammar's path has mixed field and systems work that informs — but does not institutionalize — this paper:

- Education and analytics training including a **BSc** pathway (University of London / LSE track) and an **MSc in Data Science** (University of London), plus an IBA CICT Big Data Analytics diploma
- Early exposure to **banking operations**, **agri value-chain research**, and **health-adjacent field data** workflows through internships and volunteering over several years
- Ongoing interest in how **regulated digital systems**, explainable analytics, and real-sector data quality interact
- Public writing / teaching around analytics and applied data practice

Karachi-based. Writes for practitioners who need architecture, not slogans.

Why this topic

Livestock and poultry are central to Pakistani livelihoods, yet remain hard to finance cleanly when identity, health, and price discovery do not travel together. Platforms such as Animal Passport show that **field identity** can work; capital-market rails show that **custody and price** can work. The missing piece is a **smart stack** that connects them without pretending any single institution already owns the whole design. This paper is an attempt to put that stack on one page — independently.

Contact

For critique, pilot conversations, or corrections to v1:

- Publish a public contact of your choice here before going live (email / LinkedIn).
- Prefer substantive disagreement with alternative diagrams over vague endorsements.

Author’s note on independence

> I am not writing this on behalf of any bank, hospital, ministry, exchange, university, or vendor. If I have worked with or near any organization named as a *possible* counterpart in this design, that history does not create endorsement either way. Readers should judge the architecture on its merits.

SMART LIVESTOCK ASSET STACK

Independence & disclaimer

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Independence statement

The Smart Livestock Asset Stack is authored solely by **Ammar Jamshed**.

It is **not**:

- an official publication of any employer (current or former)
- a product of any university, think tank, or research institute
- commissioned research for any client named herein
- an endorsement by Animal Passport, PAITS/PITB, FAO, AKUH, the Ministry of Health, PVARA, PSX, CDC, SECP, SBP, or any other named entity
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Named organizations appear only as **illustrative or proposed counterparties**, **public infrastructure references**, or **examples of existing field capability**.

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Forward-looking & hypothetical design

Diagrams, mint gates, RACI tables, and roadmaps are **design proposals**. They may be incomplete, wrong, or overtaken by regulation. Implementation requires separate legal counsel, regulatory engagement, and commercial contracts.

Markets & tokens

Nothing in this paper is an offer to sell, or a solicitation to buy, any security, commodity interest, or virtual asset. Past, hypothetical, or illustrative structures are not performance promises.

Third-party platforms

Descriptions of Animal Passport, PAITS, and similar systems are based on **publicly available materials** at the time of writing. Features, pricing, accuracy claims, and registration counts should be re-verified with primary sources before reliance.

Corrections

Material errors should be corrected in a numbered revision (v1.1+). Independence does not mean infallibility.

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References & further reading

References & further reading

Public starting points for v1. Expand with primary interviews in v2.

Field identity & traceability

- Animal Passport — product site
- Animal Passport — Google Play listing
- Bloom Pakistan — farmers track animals with one app
- Animal Passport — flood / paper records blog
- PITB — Pakistan Animal Identification & Traceability System (PAITS)

Market & custody rails (context)

- Pakistan Stock Exchange — public market infrastructure (engage directly for any information-product discussion)
- Central Depository Company of Pakistan — book-entry custody concepts (engage directly for any sub-account / entitlement design)

Health & policy counterparts (context)

- Aga Khan University / AKUH — referenced only as a **possible standards / credibility path**, not as a committed partner
- Ministry of Health and provincial livestock / veterinary departments — public-health and animal-health coordination

Virtual assets perimeter (context)

- PVARA and related Pakistani policy materials — confirm current legal status before any issuance design

Method note

v1 is a **synthesis and architecture** paper. It intentionally avoids fake precision on herd counts, GDP shares, or token TAMs without a cited statistical source. Add a quantitative annex when primary sector stats are locked.

SMART LIVESTOCK ASSET STACK

Glossary

Glossary

Term	Meaning in this paper
SLAS	Smart Livestock Asset Stack — the five-layer process proposed here
L0 Field capture	Apps/systems that bind animal ↔ owner in the field
Animal Passport	Biometric livestock identity / farm platform used as an L0 example
PAITS	Pakistan Animal Identification & Traceability System (PITB/FAO/GoP)
Registry company	Legal system of record for animal/batch IDs and access control
Health feed	Attested, schema-bound health record tied to a registry ID
Mint gate	Automated + legal checks required before token issuance
PVARA	Policy / regulatory perimeter referenced for virtual-asset style tokenization
Grid	Aggregation layer producing daily value feeds
Daily value feed	Published price / NAV series with methodology and verification %
CDC	Central Depository Company — custody / book-entry rail (phased)
PSX	Pakistan Stock Exchange — visibility / possible listing venue (phased)
Promote	Act of lifting a field ID into the registry company as canonical
Dual-rail	Using public PAITS keys alongside private biometric capture

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